1 **(a)** 4.5×10^5 **[B1]**

(b) 0.000308 **[B1]**

(c) 8×10^8 **[B1]**

[Total Marks: 3]

2 **(a)** 4.52×10^5 **[B1]**

(b) 0.00607 **[B1]**

(c) 1.2×10^{10} **[B2]**

If 0 scored, M1 for 12×10^9 **[M1]**

[Total Marks: 4]

3 **(a)** **(i)** 3.4×10^{-4} **[B1]**

(ii) 52000 **[B1]**

(b) 8×10^8 **[B2]**

If B0, M1 for $2 \times 4 = 8$ or $10^3 \times 10^5 = 10^8$ seen.

[Total Marks: 4]

4 **(a)** 4.52×10^{-4} **[B1]**

(b) 371000 **[B1]**

(c) 8×10^7 **[B2]**

If 0 scored, allow [M1] for 80000000 or 8×10^k where $k \neq 7$

[Total Marks: 4]

5 **(a)** $597 \times 10^{22} - 7.35 \times 10^{22}$ or $5.97 \times 10^{24} - 0.0735 \times 10^{24}$ **[M1]**

5.8965×10^{24} **[A1]**

(b) Sets up the correct division: $\dfrac{1.99 \times 10^{30}}{5.97 \times 10^{24}}$ **[M1]**

Evaluates correctly, e.g., $0.3333... \times 10^6$ **[M1]**

3.33×10^5 **[A1]**

[Total Marks: 5]

6 **(a)** 4.5×10^2 **[B3]**

If 0 scored, allow [M1] for $\dfrac{3.6}{8} \times 10^3$ or 0.45×10^3

Allow a further [M1] for correctly converting to standard form, or for seeing 450

(b) 5.7×10^8 **[B2]**

If 0 scored, allow [M1] for $(50 \times 10^7) + (7 \times 10^7)$ or 570000000 or digits 57

(c) $3.2 \times 10^{2m+n+1}$ **[B2]**

If 0 scored, allow [M1] for $(4 \times 10^{2m}) \times (8 \times 10^n)$ or $32 \times 10^{2m+n}$

[Total Marks: 7]

7 (a) 3×10^8 **[B3]**

If 0 scored, M1 for $5.4 \times 10^5 + 0.6 \times 10^5$ or $54 \times 10^4 + 6 \times 10^4$ or 6.0×10^5 seen **[M1]**

M1 for $\dfrac{\text{their } 6.0 \times 10^5}{2 \times 10^{-3}}$ **[M1]**

(b) 5×10^4 **[B2]**

If 0 scored, M1 for $\sqrt{25 \times 10^8}$ or 50000 seen **[M1]**

(c) $5 \times 10^{2x-y-1}$ **[B3]**

If 0 scored, M1 for $p^2 = 4 \times 10^{2x}$ **[M1]**

M1 for $\dfrac{4 \times 10^{2x}}{8 \times 10^y}$ or $0.5 \times 10^{2x-y}$ **[M1]**

[Total Marks: 8]

8 (a) $\dfrac{1.99 \times 10^{30}}{1.90 \times 10^{27}}$ **[M1]**

$1047.36\ldots$ or $1.047 \cdots \times 10^3$ **[A1]**

1.05×10^3 **[A1]**

(b) $(4.5 \times 10^{13}) \times 1000$ or $(4.5 \times 10^{13}) \times 10^3$ **[M1]**

4.5×10^{16} **[A1]**

[Total Marks: 5]

9 (a) 3.61×10^7 **[B2]**

If B0, M1 for $3.4 \times 10^7 + 0.21 \times 10^7$ or $34 \times 10^6 + 2.1 \times 10^6$ or 36100000 **[M1]**

(b) Converts minutes to seconds: $5 \times 60 = 300$ **[M1]**

Sets up division: $\dfrac{300}{4 \times 10^{-9}}$ **[M1]**

7.5×10^{10} **[A1]**

[Total Marks: 5]

10 (a) 6.04×10^{24} **[B2]**

If 0 scored, allow [M1] for 6.0435×10^{24} or 604.35×10^{22} or digits 604

(b) 81 **[B2]**

If 0 scored, allow [M1] for $\dfrac{5.97 \times 10^{24}}{7.35 \times 10^{22}}$ or 81.2...

(c) 1.79 **[B2]**

If 0 scored, allow [M1] for $300 \times 5.97 \times 10^{24}$ or 1.791×10^{27} or digits 179

[Total Marks: 6]

11 (a) 5.8965×10^{24} **[B2]**

If 0 scored, M1 for $5.97 \times 10^{24} - 0.0735 \times 10^{24}$ or $597 \times 10^{22} - 7.35 \times 10^{22}$ seen **[M1]**

(b) 5×10^2 or 5.00×10^2 **[B2]**

If 0 scored, M1 for $\dfrac{1.50 \times 10^8}{3.00 \times 10^5}$ or 500 **[M1]**

(c) 5.10×10^{14} **[B3]**

If 0 scored, M2 for $4 \times \pi \times (6.37 \times 10^6)^2$ **[M2]**

If 0 or M2 not scored, M1 for $(6.37 \times 10^6)^2$ or $4.057... \times 10^{13}$ **[M1]**

[Total Marks: 7]

12 (a) $\pi \times (4.5 \times 10^2)^2 \times (1.2 \times 10^3)$ **[M1]**

$7.63407\cdots \times 10^8$ **[A1]**

7.634×10^8 **[A1]**

(b) Volume in $m^3 = \dfrac{7.634 \times 10^8}{10^6} = 763.4\ m^3$ (or equivalent rate conversion $1.5 \times 10^2\ cm^3/\text{s}$) **[M1]**

Time in seconds $= \dfrac{763.4}{1.5 \times 10^{-4}} = 5.089\cdots \times 10^6$ **[M1]**

Time in hours $= \dfrac{5.089\cdots \times 10^6}{3600} = 1413.7\ldots$ **[M1]**

1.41×10^3 **[A1]**

[Total Marks: 7]